

Relatório iptables

Projeto 2 - Segurança e Confiabilidade 2025/2026- Grupo 02

1. Política por omissão

Foi aplicada a política por omissão DROP nas chains INPUT, OUTPUT e FORWARD.

A opção DROP segue o princípio de default deny: tudo é bloqueado por omissão e só é permitido o tráfego explicitamente autorizado. Esta foi a escolha mais segura para reduzir a superfície de ataque da máquina.

Foram mantidas duas exceções obrigatórias:

- tráfego de loopback (lo);
- tráfego de ligações já estabelecidas ou relacionadas (ESTABLISHED,RELATED).

2. Regras iptables

Foram usadas as seguintes redes e máquinas:

- SperServ: 10.0.0.2
- SperAdm: 10.0.0.3
- CondAdm: 10.0.3.2
- CondServ: 10.0.3.3
- Broker: 10.0.2.4
- sub-rede dos clientes: 10.0.1.0/24
- sub-rede local de SperServ: 10.0.0.0/24

As regras definidas foram:

- permitir ping a SperServ apenas a partir de SperAdm, CondAdm e da sub-rede 10.0.1.0/24;
- permitir SpertaClient apenas a partir da sub-rede 10.0.1.0/24;
- permitir ssh a SperServ apenas a partir de Broker e da sub-rede local 10.0.0.0/24;
- permitir ping de saída apenas para 10.0.0.0/24 e Broker, com limite de 4 pings por segundo;
- permitir ssh de saída apenas para CondServ.

3. Método de teste

Os testes foram executados nas webshells do Mininet-GUI. Os comandos usados foram:

- ping -c 4 <IP>
- java -jar dist/SpertaClient.jar 10.0.0.2:22345 ...
- ssh 10.0.3.3
- sudo iptables -L -v -n

O critério de validação foi simples: os casos autorizados devem funcionar, os casos não autorizados devem falhar e os contadores de iptables devem refletir o tráfego permitido.

4. Resultados dos testes

Teste	Origem	Comando	Resultado esperado	Resultado obtido
T1	SperCli1	ping 10.0.0.2	Sucesso	Sucesso
T2	SperAdm	ping 10.0.0.2	Sucesso	Sucesso
T3	CondAdm	ping 10.0.0.2	Sucesso	Sucesso
T4	Outsider	ping 10.0.0.2	Falha	Falha
T5	Broker	ping 10.0.0.2	Falha	Falha
T6	SperCli1	SpertaClient -> SpertaServer	Sucesso	Sucesso
T7	Outsider	SpertaClient -> SpertaServer	Falha	Falha
T8	Broker	ssh 10.0.0.2	Sucesso	Sucesso
T9	SperServ	ssh 10.0.3.3	Sucesso	Sucesso
T10	Outsider	ssh 10.0.0.2	Falha	Falha
T11	SperServ	ping para sub-rede local	Sucesso	Sucesso
T12	SperServ	ping 10.0.2.4	Sucesso	Sucesso
T13	SperServ	ping <Outsider>	Falha	Falha
T14	SperServ	ssh 10.0.3.3	Sucesso	Sucesso
T15	SperServ	ssh <SperCli1>	Falha	Falha

5. Capturas de ecrã

T1

```
root@SperCli1:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=63 time=5.90 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=63 time=0.516 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=63 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=63 time=0.038 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3034ms
rtt min/avg/max/mdev = 0.038/1.623/5.900/2.476 ms
```

T3

```
root@CondAdm:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=63 time=1.50 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=63 time=0.278 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=63 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=63 time=0.038 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.038/0.471/1.499/0.600 ms
```

T5

```
root@Broker:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3061ms
```

T7

```
drigo@frutasststore-password src/sperta/certs/client.keystore client-password rod
Erro de comunicacao com o servidor: Connection timed out
```

T8

```
root@mininet-v8-4 $ ssh -x root@10.0.0.2
root@10.0.0.2's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.
192 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
Last login: Fri May 15 03:08:01 2026 from 10.0.0.2
```

T2

```
root@SperAdm:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.979 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.124 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.030 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.029 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3039ms
rtt min/avg/max/mdev = 0.029/0.290/0.979/0.399 ms
```

T4

```
root@Outsider:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3066ms
```

T6

```
drigo@frutasststore-password src/sperta/certs/client.keystore client-password rod
OK-ATTEST
OK-USER

Comandos disponiveis:
CREATE <hm>          # Criar casa <hm>
ADD <userId> <hm> <s> # Adicionar <userId> casa <hm>, seo <s>
RD <hm> <s>          # Registrar dispositivo na casa <hm>, seo <s>
EC <hm> <rd> <int>    # Enviar valor <int> ao dispositivo <rd> da casa <hm>
RT <hm>              # Receber estados de todos os dispositivos da casa <hm>
RH <hm> <rd>         # Receber historico (csv) do dispositivo <rd> da casa <hm>
```

T9

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.3.3
root@10.0.3.3's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

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https://ubuntu.com/20-04

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
Last login: Fri May 15 03:30:50 2026 from 10.0.0.2
```

T10

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.0.2
ssh: connect to host 10.0.0.2 port 22: Connection timed out
```

T12

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=63 time=1.69 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=63 time=0.455 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=63 time=0.105 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=63 time=0.039 ms
--- 10.0.2.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3019ms
rtt min/avg/max/mdev = 0.039/0.572/1.692/0.665 ms
```

T14

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.3.3
root@10.0.3.3's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.
192 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
Last login: Fri May 15 02:52:01 2026 from 10.0.0.2
```

T11

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=0.621 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=1.79 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.530 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.038 ms
--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3055ms
rtt min/avg/max/mdev = 0.038/0.745/1.792/0.643 ms
```

T13

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ping -c 4 10.10.0.2
PING 10.10.0.2 (10.10.0.2) 56(84) bytes of data:gui-backend$ ping -c 4 10.10.0.2
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
--- 10.10.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3079ms
```

T15

```
root@SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ ssh root@10.0.1.3
ssh: connect to host 10.0.1.3 port 22: Connection timed out
```

Resultado de sudo iptables -L -v -n para mostrar os contadores de pacotes:

```
-n @SperServ:/home/mininet/mininet-gui/mininet-gui-backend$ sudo iptables -L -v
Chain INPUT (policy DROP 42 packets, 2616 bytes)
pkts bytes target      prot opt in      out     source            destination
 0      0 DROP        icmp -- *       *       10.0.2.0/24       0.0.0.0/0         icmptype 8
 0      0 DROP        icmp -- *       *       10.0.2.4          0.0.0.0/0         icmptype 8
 0      0 DROP        icmp -- *       *       10.0.2.0/24       0.0.0.0/0         icmptype 8
 0      0 DROP        icmp -- *       *       10.0.2.4          0.0.0.0/0         icmptype 8
46 8392 ACCEPT    all -- *       lo      *       0.0.0.0/0         0.0.0.0/0
445 54276 ACCEPT    all -- *       *       0.0.0.0/0         0.0.0.0/0         state RELATED,ESTABLISHED
 0      0 ACCEPT     icmp -- *       *       10.0.0.3          0.0.0.0/0         icmptype 8
 0      0 ACCEPT     icmp -- *       *       10.0.3.2          0.0.0.0/0         icmptype 8
 0      0 ACCEPT     icmp -- *       *       10.0.1.0/24       0.0.0.0/0         icmptype 8
 8 480 ACCEPT    tcp -- *       *       10.0.2.4          0.0.0.0/0         tcp dpt:22
 0      0 ACCEPT     tcp -- *       *       10.0.0.0/24       0.0.0.0/0         tcp dpt:22
 5 364 ACCEPT    tcp -- *       *       10.0.1.0/24       0.0.0.0/0         tcp dpt:22345

Chain FORWARD (policy DROP 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source            destination

Chain OUTPUT (policy DROP 12 packets, 796 bytes)
pkts bytes target      prot opt in      out     source            destination
46 8392 ACCEPT    all -- *       lo      *       0.0.0.0/0         0.0.0.0/0
486 52750 ACCEPT    all -- *       *       0.0.0.0/0         0.0.0.0/0         state RELATED,ESTABLISHED
 1 84 ACCEPT     icmp -- *       *       0.0.0.0/0         10.0.0.0/24       icmptype 8 limit: avg 4/sec burst 4
 1 84 ACCEPT     icmp -- *       *       0.0.0.0/0         10.0.2.4          icmptype 8 limit: avg 4/sec burst 4
 4 240 ACCEPT    tcp -- *       *       0.0.0.0/0         10.0.3.3          tcp dpt:22
```

6. Conclusão

A firewall foi configurada corretamente segundo uma política de default deny. A máquina SperServ só aceita o tráfego pedido no enunciado e bloqueia o restante por omissão. Os testes realizados confirmam o funcionamento esperado.